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Designing for Awareness: Applying Friction in Dark Patterns

An exploration of design friction in promoting critical thinking within front-end interfaces.

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Abstract

A prevalent design practice in modern user interfaces is the deployment of deceptive 'dark patterns', which coerce user behaviour in favour of the organisation. Dark patterns describe manipulative design practices such as vague phrasing or hidden information designed to deceive the user. These coercive practices can lead to various leaks of private details occurring which gets spread around to third-party organisation. This project aims to explore how design friction can be used in front-end design to promote critical thinking towards dark patterns. By adding intentional resistances within the interaction, it has the potential to make users more aware of their decision-making. Following a research through design (RtD) methodology, a series of five prototypes were developed and served to collect qualitative data. The results demonstrate that design friction within front-end design has the potential to increase user awareness depending on the amount deployed.

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1 Introduction

Modern front-end designs prioritize frictionless interactions, where interactions such as cookie banners are instantaneous and thoughtless. Most interactions are meant to provide the user with a sense of ease through the interface as to make the overall website look straightforward. Whilst these interfaces are sleek in their design, behind the scenes there are several systems that do not expose their actions to the user. These actions such as accepting privacy policies or sending personal information has several systems which are not visible to the user, making the result unnoticed or not thought of.

Khanan et al. (2019) explains how the operation behind collecting private information has exponentially increased over the years. Furthermore, they elaborate on the fact how the most sensitive problem with data is it concerns against privacy. While social media and health informatics provide immense datasets, the lack of regulatory oversight regarding how organizations collect this information remains a critical vulnerability. This can lead to unregulated collection of information by different organisation. Despite the apparent simplicity, they often incorporate strategic design patterns that lead users to perform unintended actions, effectively bypassing critical thinking. According to Mathur et al. (2019), these strategies are known as dark patterns and are prevalent throughout many different websites. Dark patterns are used to coerce users into choices that benefit the online service rather than themselves. They are deceptive designs that are common when browsing through various websites such as online shops, social media, or news outlets. Dark patterns are becoming increasingly more common across various websites and regardless of the kind, their end goal is to deceit users. As previously mentioned, cookie banners are one such example of dark patterns where they purposefully set an abundance of information as to let the user more hastily decide. As seen on figure 1, other examples include arbitrary choices such as some buttons giving the option of “Reject Non-Essential” without any context as to what is essential. Mathur et al. (2019) explains that third-party organisations are often responsible for dark patterns on behalf of the websites. Consequently, these deceptive practices are significantly more pervasive across the web than users generally anticipate.



Figure 1. Example showcasing information overload and vague wording. Note. From Vox Website by Vox Media, Inc. (2020)

1.1 Research Question

Due to the added risk of coercing users into performing tasks which are not desired, there is a distinct lack of critical thinking that these dark patterns persuade users from. Distler et al. (2020) highlight how a lack of critical thinking in user interfaces leads to daily security risks, suggesting that design friction could be a potential solution to enhance user awareness. In contrast to dark patterns, friction can help to promote reflection during the interaction due to the slower design aspect.

This poses the question of how can friction in front-end interfaces be used to create more awareness amongst users? Could slower interaction within front-end design potentially promote critical thinking as to counteract certain dark patterns? By introducing intentional resistances within the interaction, it could possibly shift user experience from simple clicking to a state of awareness regarding data privacy.

2 Background

2.1 Data Brokerage

The mechanisms governing the transfers of personal information is unknown as it is often obscured by layers of third-party involvement. Furthermore, while individuals acknowledge the inherent value of their personal information, there is a significant lack of awareness regarding the broader implications of data sharing. Additionally, the full extent of data transfer remains vague, however, it is significantly vaster than anticipated. Reviglio (2022) highlights that while personal data is an economic asset, the politics of the industry remain unknown. Due to the vast scale of the operations and how wealthy the industry is, consumers are unaware of the specific processes by which brokers aggregate and sell their information. They elaborate that due to the increase of digital consumption this trend will increase and lead to more personal information being sold by third-party organisation. Moreover, the data brokerage industry is complex as it could be several different companies that run it regardless of being data brokers. For example, various organisation such as internet companies, advertisers, retailers, and trade associations sells personal data in various forms and could therefore be considered data brokers despite their original intent (Reviglio 2022; Pinchot et al. 2018; Crain 2018). According to Pinchot et al. (2018), companies purchase information which include names, email addresses, interests, and offline activity about their prospective customers. They note that, all data does not necessarily have to be collected from private sources rather it can also be public records such as consumer purchase data and other web browsing activities that can be scrapped. Reviglio (2022) explains that

consumers are more susceptible to data transferring through social media and that data brokers' market is concerned with marketing and online advertisement. This helps sellers to understand what customers want, thereby delivering marketing messages that more relevant and consistent. Despite there being tools for help concerning the collection, it still does not mean full transparency across various platforms. According to Crain (2018), regardless of what data is collected, individuals' perception of these operations are not widely known, leading to a privacy asymmetry. It has become accepted for these practices, which in turn has scrutinized the critical thinking towards those operations. They note that data broker industry has operated without public awareness or direct regulatory oversight. This demonstrates a lack of transparency regarding user actions, which fails to communicate the subsequent consequences or the potential for data to proliferate among third parties.

Deulkar & Gupta (2018) explains that even though individuals tend to reveal details about their personal life without realizing, the collected data can also be used to make guesses about a person and their interests based on the received information. In addition, websites give away majority of information about the user as certain sites use cookies to track user's trails.

2.1.1 Repackaging Data

Anthes (2014) notes that data brokers collect, analyse, and repackage consumer information for sale to external companies. Subsequently, companies dependent on this repackaged information purchase the data for their own operations which can be used for marketing efforts or to create references for individuals. According to Ijuo (2025), the valuable commodity of data forms the groundwork for future analytics and predictive modelling. Moreover, higher volumes of data only increase the potential for accurate predictions that they can tailor for consumers in commercial offerings, which, helps business competitive advantage and profit. They note that data brokers serve as intermediaries in the data economy, therefore they are not necessarily responsible for what the data is used for. That said, they enable the seamless transfer of personal data which comes prepackaged with concerning the company's needs.

2.1.2 Restructuring the Operations

As more platforms and application get increasingly larger, so does the collection of personal information. Given the vast scale of industry operations, there are no signs of a reduction in this trend. That said, Bajwa & Meem (2024) propose a framework for restructuring these operations, specifically through the integration of privacy operations into data brokerage systems from the beginning. They explain that through incorporating data protection as early as possible, data brokers are to consider privacy during the design stage, which could ensure that personal data is processed with the highest privacy settings. Additionally, as the roles and definition of what

could be considered data brokers, it should be defined within their organization. For instance, they argue that defining specific roles within the data aggregation process facilitates the mapping of data flows and the assessment of risks involved. In addition, establishing clear consent mechanisms for data collection is a crucial factor in ensuring that consent is freely given and specific to what it is about. Websites should create transparent mechanisms that make it simple for people to give and withdraw consent.

According to Kanwal & Walby (2014), they argue that data brokers claim to work with anonymous data, however, since there is multiple exchanging information at once means it is hard to track accountability. Data subjects are entitled to know what personal information of theirs is being processed and how. Additionally, they argue that this can help with showing clearly if personal information is being processed by the controller or by a third-party processor. These measures aim to provide users with a clear distinction between data practices, prioritizing transparency over the obfuscation of corporate objectives. These propositions reflect the structure rather than proposing solutions on how it should be changed. Due to the sheer size of the industry surrounding this, trying to propose solutions presents significant challenges rendering any such effort inherently ambitious.

2.2 Dark Patterns

While personal information is often collected through accessible sources like public records and email addresses, many websites deliberately employ deceptive design patterns to manipulate users into performing desired actions by the company. These are known as *'dark patterns'* and is widely used across several different websites. The term is used to describe malicious design strategies that coerce users into performing actions that benefit the respective service rather than themselves (Gray et al. 2018, Schäfer 2024). As of recent years, these strategies have exponentially grown and are being purposefully used in various service websites, as well as social media platforms. Gray et al. (2018) elaborates that even though persuasive design is often good as it can be capable of encouraging socially responsible behaviour, there is substantial ethical considerations regarding designing interactions that involve coercion. Dark patterns are not necessarily primarily focused on personal information, as it contains several design patterns that can occur in various forms.

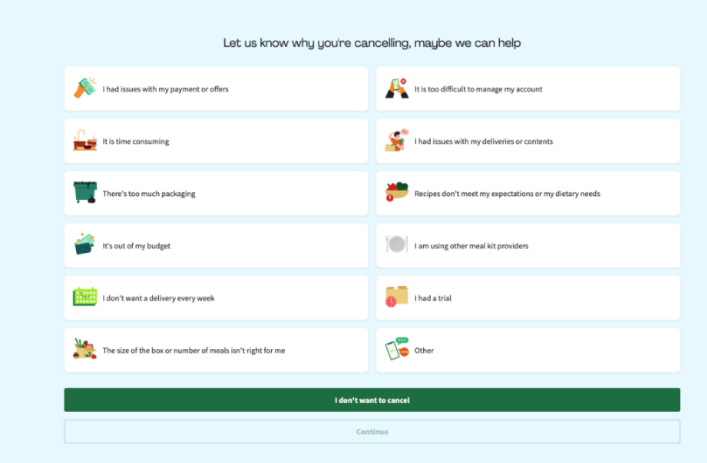


Figure 2. The cancel subscription page from Hello Fresh where the continue button is purposefully obscured below with an overwhelming amount of option to describe why you are cancelling Note. From Hello Fresh Website, by HelloFresh SE (2025)

Gray et al. (2018) outlined how dark patterns come in diverse ways, for example, websites that hide additional information concerning a specific product is in itself a dark pattern as they purposefully hide valuable information from the consumer (see Figure 2). Moreover, they highlight another example of dark patterns where the website adds items not chosen by the consumer to their online shopping cart, claiming it to be a suggestion based on other purchases. They use the definition as “interface interference” which is meant to confuse the user or limit their discoverability of potentially important actions (p.7). Furthermore, hidden information and aesthetic manipulation tend to be on websites that involve the action of forcing the user to create an account to be able to perform tasks. Gray et al. (2018) explains that aesthetic manipulation is a design method used for any user interface that deals more directly with form than function. This includes choices such as making the user’s focus on one thing to distract them from or convince them of something else. They explain that aesthetic manipulation plays a significant role in dark patterns as it possesses several qualities in which can be made to deceive users. In essence, regardless of the back-end of a website dark pattern are meant to change the users’ perception on the user interface, deceiving them into thinking there are no bad decisions.

According to Luguri & Strahilevitz (2021), to manipulate and individual to a certain choice is to subvert their decision-making and to deny them authorship over their decisions. Manipulation is meant to be false beliefs on the user that make them believe they are making the right choice. Dark patterns inherit this quality, as it is a design strategy meant for success from the creator’s side. They explain in their study that dark patterns contain several attributes, suffice to say, they possess taxonomies which follow

various design choices all of which are manipulation. Furthermore, these design choices could potentially have unanticipated societal consequences as interfaces that make users turn over personal data could make databases create more detailed descriptions on individuals for advertising purposes. As previously mentioned, data brokerage is a large industry containing of several companies behind it, which is why dark patterns plays an influential role in their operation.

2.2.1 Deceptive Data Commodification

Luguri & Strahilevitz (2021) highlights an example where Facebook harvested user data in an effort to impact the U.S. presidential election in 2016. These deceptive dark patterns fail to disclose how individual preferences are harvested to influence future behaviour leading to significant and unanticipated consequences for the user. Regardless of how anonymous users are in a website, companies hide that through the front-end design which does not expose the several mechanisms in the back-end. As mentioned by Reviglio (2022) data holds political value as it forms an individual's preferences which could potentially be used for beneficial use. It highlights dark patterns are meant to deceive or manipulate users, they can do so in a manner that hides their core mechanics.

There are several ways in which data collection occurs, whether that be through consent forms or account creation. Nouwens et al. (2020) elaborates that an important aspect of data collection is through consent in relation to a particular and specific purpose of processing data. They argue that regardless of linking to an external list of potential vendors does not represent the actual code being ran in the background and thus is insufficient information. Consent should be able to be rejected and not displayed through arbitrary choices that are not explicit enough. Nouwens et al. (2020) highlights in their study consent forms in which contain common dark patterns tropes that have been established. In their study they concluded that implicit consent is common amongst privacy consent forms, most of which made rejecting more difficult than accepting, as well as sites having pre-ticked options already checked despite no input. Furthermore, consent can never be valid in the presence of various third-parties involved, making the consent forms hard to adjust to since you are forced to design around their needs (Nouwens et al. 2020; Mathur et al. 2019). Their study reveals that participants who were apart of the consent forms did not care for the repercussions. Study participants indicated that they primarily engaged with consent forms with the sole intention of completing the process as quickly as possible. This results in dark patterns removing the notion of critical thinking or any realisation of consequences being made during the process. It showcases how dark patterns are usually managed through pragmatic reasons rather than any critical ones as they are seen as nuisances that should be dealt with quickly. However, Nouwens et al. (2020) highlight a juxtaposition in user behaviour, while participants often did not care for the consent forms, they

simultaneously express significant concerns regarding their digital privacy. As previously mentioned, this would explain how regardless of users' perception is on their data collection, taking the steps to oppose those aspects are usually not acknowledged.

2.2.2 Bright Patterns

Despite dark patterns are perceived as deceptive and manipulative there are circumstances where manipulation is not inherently negative. According to Susser et al. (2019), manipulation can occur throughout our daily lives that does not necessarily dictate that they, for example, friends that share social media posts about health concerns on their profile manipulates our minds into taking better care for ourselves. They elaborate that manipulation cannot simply mean to make less-ideal decisions since there are positive ways to approach it. This contrast is important to note for dark patterns since they work around the notion of negative manipulation.

2.3 Design Friction

It is important to note that the mention of design friction is primarily focused on designers, which in the case of this are to create more user-oriented designs that aid users not to perform hasty decisions. The responsibility of the designer is to ensure that data sensitive interactions are deliberate rather than hastily done. While the analysed dark patterns succeed in persuasion, they fail to provide the cognitive space required for critical thinking. As data brokerage can dictate the design for front-end interfaces, friction is only meant as way to apply resistances within the interaction as to make users more conscious of these dark patterns.

Mejtoft et al. (2019) conducted a study on using design friction as a means of forming better understanding concerning various interaction. They define friction as a means of creating hindrances towards user's from easily achieving the goal. Furthermore, the authors argue that the primary drawback of friction is the notion of mindless interaction. Their study concludes that introducing intentional friction can deepen a user's comprehension of certain information. They conclude that there is a correlation between friction and higher levels of satisfaction after an interaction. As dark patterns concern themselves with information overload and confusing users from the interaction, friction could potentially be used as a means of deepening critical thinking towards these issues. As previously mentioned, dark patterns tend to be seen as nuisances rather than interactions that are meant to be seemed as thoughtful. Because of this, dark patterns lack any sort of resistances within the interactions to allow for understanding of what consequences could potentially entail (Elteren 2020). Suffice to say, by intentionally including friction as means of stopping users to form awareness could potentially help with creating an understanding. This can facilitate better comprehension of the task without necessarily

abandoning the underlying intent of the original intention of banners for example. Elteren (2020) proposes that consent should be understood as an action, not as a subjective phenomenon. Additionally, consent should rely on the giver to be in control of their decision, meaning that the consent is given freely.

Cox (2016) uses the term 'microboundries' to describe friction as a means of reflection during the interaction (p.1392). Microboundry is meant to provide stops during the interaction as to make the user not rush from one context to another. Despite that dark patterns are meant to nudge users into behaviours that are not desired, adding stops are meant to create periods of reflection during interaction. They explain that microboundries can be created by adding additional steps into interaction. As previously mentioned, friction can be done throughout various things such as off-putting colouring or visuals, but the addition of stepwise interaction could potentially be suitable in the case of dark patterns. Cox (2016) mentions video games as an example of friction as they involve several steps where users are challenged through different types of breakdowns. As Kocyigit et al. (2024) observe, there is no single attribute that consistently defines dark patterns. This lack of a unified definition complicates efforts to address the issue, as the interactions involved are inherently multifaceted. Through their study they propose the argument that to counteract these issues concerning dark patterns, there should be symmetry between the negotiator and the user. In essence, it is meant to form a fair response between the two parties which could potentially make for objective decisions to be made. Although they do not explicitly mention friction, they argue for consent interactions that involve restraint, effectively prioritizing the interface's restrictive qualities to ensure user engagement.

As mentioned earlier, the point of modern front-end design is to be as efficient as possible, without causing any problems during the interaction. Benedetti & Mauri (2023) explains how in contrast to smoothness in design, friction is meant to identify the user interface and user experience that require effort to achieve a goal. In addition, they mention the term frictional interfaces as opposed to frictionless interfaces that are meant to recontextualize existing to data to challenge its initial context. They conclude in their study that friction is most fitting when describing tackling of data, interfaces, and algorithms. Although friction is applicable across many interaction contexts, its most critical purpose lies in the handling of personal data, where it serves as a necessary safeguard against mindless disclosure.

2.3.1 Visual Clarity

In the context of dark patterns, many manipulative strategies rely on visual deception to mislead users. As previously mentioned, these tactics are information overload, and this forms a concern as it leaves a lot of crucial detail left unread. More importantly, since a lot of information is not being

properly understood, it makes the friction more of an important aspect. Otten et al. (2020) argues that good aesthetics helps to improve user experience and useability. They define it as 'visual clarity,' which is a concept based on the perception of design elements (p.11). They argue that the connection between aesthetics and useability has a considerable influence on users' perceptions of user interfaces. These aesthetics are manipulated by many factors and form the eventual performance for how the interaction may follow. Otten et al. (2020) explains that complexity and order are shown to have significant impact on user efficiency and lack of understanding. Therefore, to improve the overall experience visual clarity is meant to form balance between user impressions and overall sophisticated approach. They state whilst visual complexity overall helps to improve the time required to visually orient oneself, consequently it makes it harder to learn and makes the interface overwhelming. In the context of design friction, time is not as an important as the concept revolves around resistances within the interaction. That said, if a user interface has a clear and well-structured impression it can help to positively influence the assessment of usability.

2.4 Related Works

Several works concerning the question of how friction can potentially be implemented within front-end design has been published by many researchers.

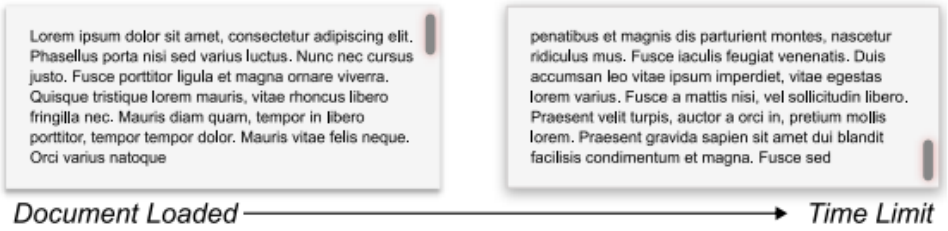


Figure 3. Image from their project in which they showcase how the page scrolls automatically depending on the time limit Note. From User Interface Constraints to Influence User Behaviour when Reading and Writing (p. 3), by Joshi (2023).

In Joshi's (2023) research project, they conducted a study in which followed user interface constraints to influence user behaviour reading and writing. The project was focused on applying constraints on reading habits that involved putting sentences followed in succession as to see what influences could be caused by such interactions. Moreover, they note it as 'time-bounded reading' where the document would automatically scroll down to encourage the reader to read at a specific rate. It was meant to enforce a certain pace and force the user to read the sentence instead of skimming through (see Figure 3). Whilst friction is not explicitly said, their project holds merit in those regards either way. As previously mentioned, dark patterns involve

information overload as tactic to persuade users which leaves the sentences barely read with no understanding as to what actions effect you.

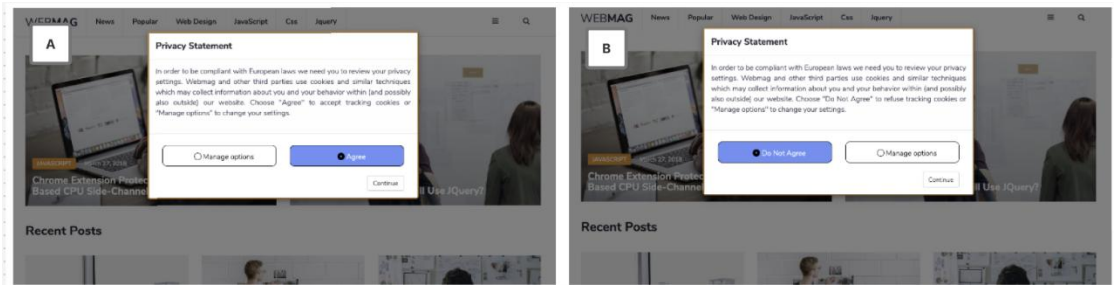


Figure 4. A. Shows how initially dark patterns are produced wherein 'agree' is brightly displayed and there are no options of simply disagreeing otherwise the user must select 'manage options' B. An altered version which 'Do not agree' is highlighted and 'manage options' is the other selection. Note. from Dark and Bright Patterns in Cookie Consent Requests (pp.9-19), by Graßl (2025)

Graßl et al. (2025) conducted research which involved design nudges as means to influence user behaviour and perception within the context of cookie consent requests (see Figure 4). Their project involved at subverting dark patterns and implementing bright patterns that steer towards privacy-friendly options. In addition, they conclude that participants expressed favourably for obstructing a choice led to a higher sense of control. As previously mentioned, consent forms follow a yes or no direction, in which they remove tediousness as a means of the user thoughtlessly choosing an option hastily. The work of Graßl et al. (2025) is an important consideration here, as reducing the number of choices often helps convey the most critical information to the user more effectively.

Jacobs & Schraffenberger (2020) did a project related to design friction for privacy, wherein they implemented intentional stops within the interaction as to form forced reflection upon users. Their research explains how usual user experience concerning front-end are more concerned on intuitiveness and efficiency. In contrast, this leaves potential data more vulnerable to various third-parties, hence why implementing friction amongst data sharing could potentially help solve this and form reflection. In essence, their project reveals the fact that friction has the potential to affect users when interacting with sensitive data. It is also important to note that this project highlights the power of friction. It shows that even in privacy-related contexts, adding obstacles can steer a user's actions, even if it makes the process less efficient.

3 Methodology

3.1 Interviews & Research Gathering

Before work began on the initial design process, interviews were conducted as a means of research gathering. Following Kvale (2007), they explained that interviews should encourage the interviewees to describe the points of view on their lives and worlds. Furthermore, they elaborate that interviews are usually prepared with a script that will function as guidance for questions, for which this part of the project heavily followed accordingly. Kvale (2007) states that certain studies will demand a more structured and predetermined sequence of questions rather than more spontaneous and quick judgements.

Each interview was audio-recorded and later transcribed. These transcripts provided the qualitative foundation needed to analyse participant perceptions and understanding of deceptive dark patterns. The interviews served as the foundational phase of the overall design process and would not be revisited that much once prototyping would begin. That said, the research gathered would be used extensively throughout the whole project.

3.2 Research Through Design

This project follows a research through design (RtD) approach as ones of its main methods. According to Dalsgaard (2010) article concerning the approach, research through design can be defined as focused on improving interaction design practices and theory, using the act of making as a primary tool for research. They elaborate on the reasoning to choose this as it pertains to finding particulars of a given situation rather than generalizable findings.

As dark patterns are quite a large topic since it involves many distinct aspects of front-end, the priority lies in uncovering the underlying mechanisms that promote reflection, rather than attempting to propose a single design intervention. With this in mind, research through design takes a designerly approach that is both iterative and explorative which in the topic of dark patterns helps since there are several approaches that can be used as to help circumvent the issue.

The project adopted a research through design approach, as the goal was not to produce a definitive solution, but rather to investigate how user interfaces are constructed and perceived. For this reason, the prototypes being made was meant as the primary ingredient for gathering research rather than trying to propose solutions.

3.3 Prototyping

The purpose of this project is to create several digital prototypes that pertain to various interaction within promoting reflection. Houde & Hill (1997)

explains that prototypes are meant to represent future ideas and concepts rather than direct answers pertaining to the project. They elaborate that by focusing on the purpose of what the prototype is can entail to better decision making. The project heavily follows the notion prototyping as a means of understanding the concept behind potential reflective user interfaces. Wensveen & Matthews (2014) explains the various roles of prototyping wherein the third role they elaborate that prototypes can refer to 'research archetypes' (p.268). In essence, it elaborates how research archetypes have physical embodiments of concepts, wherein they form understanding in the space in which they occupy. Even though their example is within physical examples of prototyping, it still holds merit within digital prototyping as well.

3.4 User Testing

The user tests are an evaluative phase, allowing participants to interact with the prototypes to gather qualitative data on the design's effectiveness. According to Bastien (2008), user-based evaluation is a procedure meant for users to explore and have their behaviour observed. Moreover, they explain that regardless of the interactive system being towards software or a website, the primary goal is to assess the effectiveness of the prototype.

During each user test, comprehensive field notes were collected to accurately document participant responses. Participants opinions and thoughts would be written down along with the user testing. This qualitative data was gathered primarily for analysis purposes, which could then be used to form thematic analysis. The field notes were documented in a way that captured any significant words being used to describe their thoughts and did not contain any transcripts or audio-recordings.

Quantitative methods would also be deployed as the project would introduce a rating scale to be used as an evaluation method with each prototype. This was mainly used as a means of gathering an overall estimate of each prototype. Moreover, these findings can be used to map a comprehensive overview of where each prototype sits in relation to one another. After the participants would finish the prototype, they would then note down a mark in which they believe it lies within the scale. The scales would then be layered with all the other ones to get an estimate of all the participants notes.

3.5 Data Analysis

This study utilizes thematic analysis to evaluate the gathered information from user testing. It is important to note that this method only pertains to the things received from user tests and not the previous interviews. According to Braun & Clarke (2006), a theme serves to isolate key patterns of meaning within a dataset that directly intersect with the overarching research question. This is meant to represent some level of patterned response or meaning with the data set. Furthermore, Braun & Clarke (2006) note that themes can be identified using either an inductive or a deductive approach.

While an inductive approach allows themes to emerge organically from the raw data, a deductive approach relies on a pre-existing theoretical work to guide the analysis. This project heavily utilizes a deductive approach, relying on the theoretical concepts established in the background to categorize and analyse the empirical data.

Afterwards, all of the gathered field notes would then be coupled with the themes within an affinity diagram. There, everything that has been said will be connected to themes that emerged from the literature along with the field notes to establish a connection between the themes and field notes. The affinity diagram was made to visually organise and analyse the qualitative insights gathered from the user tests

3.6 GDPR

None of the participants had their information stored within a collection, nor was it ever shared. All of the participants that have contributed to the research will not have their name shared as to remain anonymous. Before the interviews and user tests were conducted, participants were asked and given a form for their consent to be recorded in compliance with General Data Protection Regulation (GDPR) policies. In addition, all recordings were stored locally on the computer with nothing being stored on an online cloud. The prototypes themselves had no way of storing data and would automatically delete any written information after the interaction.

4 Design Process

4.1 Interviews & Research

Initial steps for the project pertained to research gathering and interviews. The purpose of this was to gather the groundwork for how dark patterns are more prevalent than we think. As most individuals are aware of what happens in the back-end concerning data collection, not everyone is aware that it is happening in the interface. Because of this, structured interviews were conducted to gather diverse perspectives on user experiences with persuasive front-end design. A total of four participants were interviewed with 10-15 minutes in total. The interviews were structured objectively to focus exclusively on exploring the core topic and uncovering participant experiences rather than gathering subjective opinions. One common trope that would be brought up on each discussion was the notion of cookie consent being seen as malicious. One participant notes *“Let's take an example. I go to Elgiganten, you can log in and they ask about cookies and all that. I would rather not....”* It is important to note that none of the questions disclose anything specific to cookies, but three out four participants brought it up

themselves. This would align with the earlier research done about cookie consent forms being seen as nuisances and nothing that individuals particularly show favourable to. More importantly, this same participant shared how they are not aware of what information spreads, saying "*I do not know how it spreads. They trick you. You do not know what they do with the information and you do not see the difference when you log in.*" As previously mentioned, while most individuals are aware that data sharing occurs, the underlying processes remain largely unknown. This lack of a reference point makes it difficult for users to have any meaningful reflection. When the process remains obscure, it becomes much harder for users to critically evaluate the reasoning behind it. One participant expressed uncertainty regarding the notion of reading policies stating, "*No. I never read through them. I do not think anyone does that. I do not understand anything there."* Due to the information overload that most consent forms have, they overwhelm users with the quantity of text that does not register accordingly to everyone. That same participant goes onto elaborate that they simply accept how it is and throw the information away from their mind. One of the core issues with standard consent interactions, such as cookie banners, is the overwhelming abundance of information they present. This tends to lead to fatigue amongst users as evident by the participant remarks concerning it.

The common consensus on dark patterns is that they are misleading about specific choices. Therefore, a showcase of the back-end could potentially prove feasible as it helps to establish its inner workings. A proper balance must be set up so that users understand the potential risks of their data, which typically involves its use in data brokerage. In all the interviews conducted, none of them understood why organisations sell their data to third parties. Only one participant states "*Quite generally… You constantly see someone coming out with some new data breach. Where they have found information that ex-companies should not have had.*" Regardless, the concept is still unknown to them, thus, this suggests that the concept needs to be highlighted somehow within the interaction. The design opportunity with friction is here to provide the user with sensible information that makes them reflect over the potential consequences an action has. Despite friction being used as a means of resistance within the interaction, the problem is how to provide that information without causing fatigue among the users. When participants were asked about if their online decisions were based on themselves rather than being guided, most participants said no, however felt that it is algorithms that steer that. One such participant states "*Both, actually. For example, if we take the feed on social media…*" Ultimately, guidance within digital contexts is beneficial provided it does not prioritize algorithmic values over user intent.

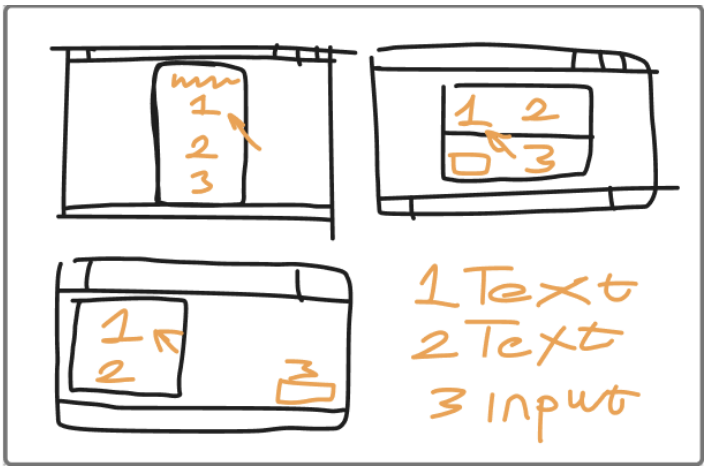


Figure 5. Quick sketches following a layout for stepwise interaction involving time-bounded text

Since all the perceived information goes unnoticed means that there are potential design opportunities concerning how to convey text in a more sensible manner. Following Joshi's (2023) project on time-bounded text, it helps provide context as to how to approach the issue in a way that could possibly convey to users more understandably (see Figure 5).

Joshi (2023) proposed an auto scrolling feature for text, likewise, the sketch is meant to represent that same idea. However, in this case the user must take an active decision to highlight parts of the paragraph, in which it will slowly read out. This stepwise interaction is designed to enforce reading by preventing the user from simply skimming through all the information at once. Compared to the auto scrolling of text, this will involve the user making conscious decisions to highlight the text that they are about to read. In contrast, this would simply make the interaction completely time-bound which in this case does not seem feasible as it is closer to a waiting for the time to tick down. The prototype incentivizes users to read by requiring them to actively engage with the text through the deliberate highlighting of each individual paragraph. As mentioned previously by one of the participants, cookie banners have an abundance of text which most users tend to skim through. The aim with every prototype is to keep every description as brief as possible, as to not deter users with endless reading about certain policies. More importantly, the descriptions should help to convey what it is trying to do and how it important for a private person. That said, rather than providing an in-depth description about every detail about data brokering, each description will explain how their data is being transferred with an additional text about outcomes as well. Kanwal & Walby (2014), explained how there are multiple exchanges of data going on at once which the prototypes are meant to represent by highlighting the potential oversharing through various third-party organizations. As highlighted previously, the aim is not to propose solutions as to how fix the collecting of personal data, however, instead try to

analyse potential design opportunities that help promote reflection upon the action we take concerning dark patterns.

It is difficult to decide the optimal amount of information to provide, as the negative belief of this topic during interviews suggests that the prototypes may be viewed critically from the start. The prejudices concerning consent forms within in cookie banners are not perceived as favourable due to their nature as evident by one of the participants stating, "*I take the time to turn them all off, one by one - Then there are necessary cookies, whatever that is....*" This concern is crucial because it makes it difficult to determine whether perceptions of the topic are objective or if they stem primarily from existing disdain. Likewise, it as well makes the topic surrounding this quite sensitive from the starts since it works around the notion of personal data. The aim is to understand the current situation and identify the opportunities that arise from them.

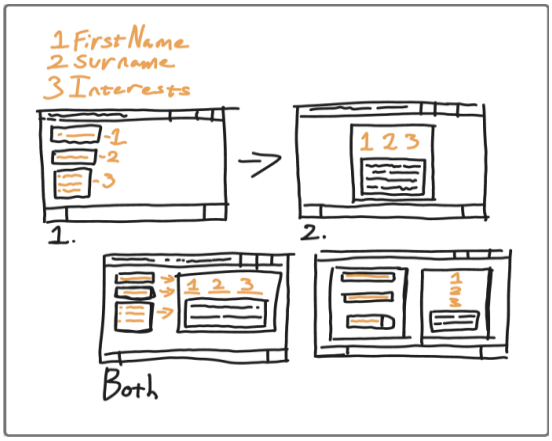


Figure 6. Sketching potential layout designs for the prototypes

4.1.1 Post-Interview & Ideation

After interviews were conducted, it became evident that regardless of the perception on dark patterns there would be a lot of ways in which it can be explored. Not only does the issue of information come at play, but the interaction itself is important since despite some dark patterns making it so users take a conscious decision for something, it does not mean it inherently is well. For example, one participant said *"I was going to order fast food via Foodora. I started a notification so you can see when your order is coming. - And then I was spammed with notifications about other deals"* which helps to highlight how persuasive these interactions are as they come off too discrete but enough for the user to accept additional steps due to the context. Consequently, sketches were used to quickly block out the shapes where information would be stored as to determine the exact points where users should read (see Figure 6).

Finding a good balance between informing the user whilst not causing fatigue within the interaction was a challenging task. Though design friction offers the possibility of reflection, the perceived nuisances on dark patterns make the information part particularly troublesome. While the initial ideation felt stagnant, the process eventually shifted towards developing a diverse set of interaction within design friction against dark patterns, each utilizing varying degrees of interaction. Since the term dark patterns involves many different deceptive user interfaces within front-end, the future prototypes developed will span a spectrum of engagement, ranging from designs that prioritize forced reading to more gimmicky interactions, to identify exactly where the line is drawn between effective critical thinking and user frustration. This choice was intended to demonstrate that despite the negative perceptions surrounding data collection, information can be delivered through active engagement rather than rigid interactions. Evaluation reveals that participants' feelings of being cheated often stemmed from a lack of perceived effort in their interactions. The purpose of design friction is to combat this by introducing intentional resistance to ensure that users do not complete their task swiftly. Hastily skimming through interactions within dark patterns are a core issue in which leads to benefit the organization rather than the individual. As previously mentioned, design friction is not intended to prolong an interaction and exacerbate an existing nuisance, but rather to intentionally steer users' effort into something more considerate.

4.2 Prototyping

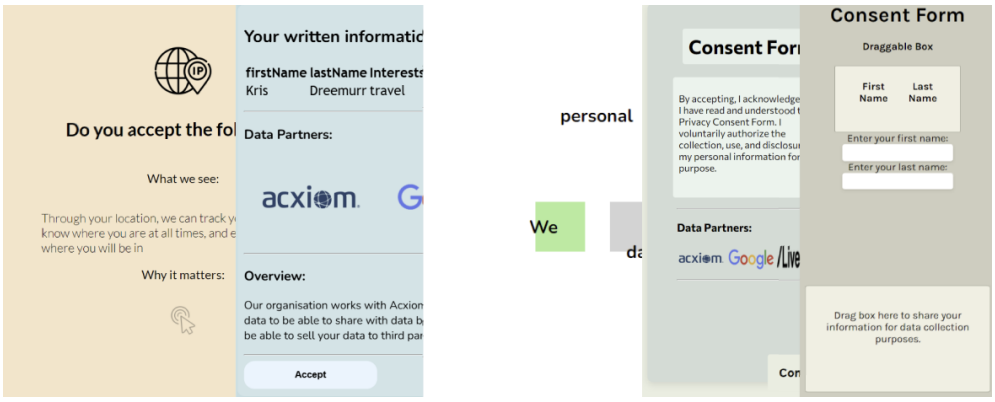


Figure 7. Collage of all five prototypes that were developed over the course of the project

Once the interviews and ideation were finished, the process shifted swiftly to the prototyping phase, as this stage required the most considerable time investment for the project's development. Each prototype would contain a variety of interactions that stemmed from simple to more exaggerated. All prototypes were built from the ground up through HTML, CSS, and JavaScript, with each of them being made separately and made from scratch. As shown on Figure 7, a total of five prototypes were made with one of them being made specifically for smartphone interaction. Some prototypes follow

are more rigid approach with it being more approachable, whilst other prototypes

exaggerate a bit on the usual interaction to gauge where the line between gimmicky and critical thinking draws. Additionally, since data collection occurs through variety of interactions, the prototypes explored various themes to highlight the vast and diverse nature of these dark patterns.

It is important to note that none of these prototypes are intended to be definitive solutions for online security. Rather, they serve primarily as a design opportunity to explore friction within front-end interaction and gather information on the potential critical thinking they can provide. This project took a Research through Design approach, which lead to a series of successive prototyping cycles. Each iteration was purposefully built to explore a unique set of front-end interactions and evaluate its impact on user engagement. The following section outlines the development of each prototype about how they function and what their results were.

4.3 Conducting the User Tests

Once each prototype was finished, work transitioned into the user testing phase. The intention was to have the participants go through each prototype and give a brief overview of their experience and thoughts. It is important to note that none of the participants were ever guided through their experience or provided a description of what each entail. This was to keep the participants thoughts as unique as possible and maintain a neutral stance towards these interactions. Furthermore, the specific scope of the project was never explicitly disclosed. These types of data interactions occur daily while browsing the web, sometimes without our own knowledge. Because of that, it felt most appropriate for the user tests to reflect those archetypes. By leaning into that narrative, the study aimed to elicit more spontaneous and authentic

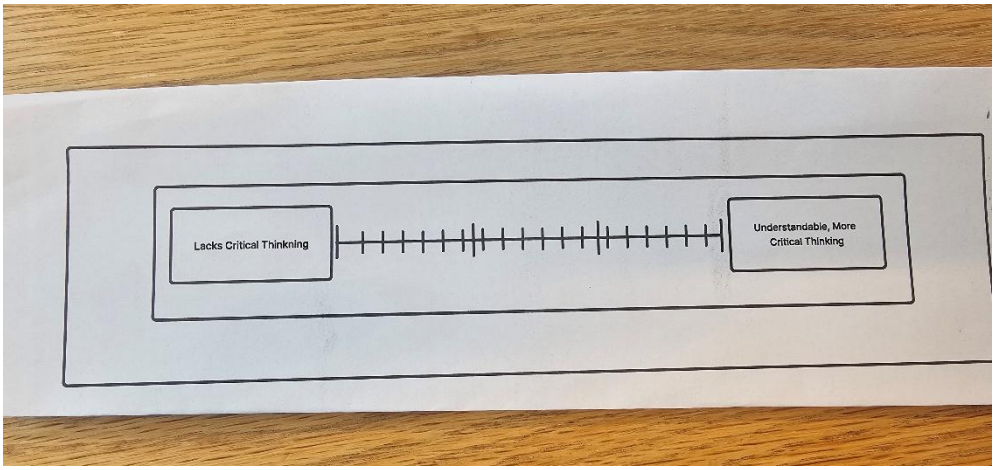


Figure 8. The scale used in which each participant would mark where they think the prototype should be placed

after everything was finished as then it could potentially generate more general thoughts surrounding the topic.

Instead of merely making the participants quickly skim through the interaction, a scale was made to gather additional information (see Figure 8). In hindsight, the scale comes off as quite arbitrary with its wording, however, it helped to produce more thoughts as the participants had a way to jot down their thoughts. Once the prototype was finished the participant would then mark on the scale to which extent they think it helped promote critical thinking. The scale goes from 'Lacks critical thinking' to 'Understandable, more critical thinking,' point being that depending on their opinion or experience, they would mark where the prototype fits in those regards. If the prototype was too plain or gimmicky, then it would be appropriately within the left side and vice versa. It is important to note that there was no real metric for scale to count for, so it was merely judged by the placement and the words from the participant.

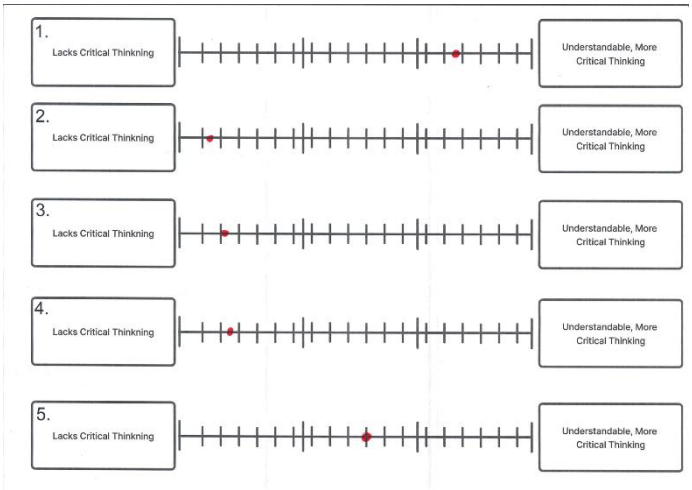


Figure 9. A summary of one of the participants where all five were scanned and then later put together.

After testing all the prototypes were finished, participant experiences were mapped onto an overall scale to provide a comprehensive overview of how different design faired against each other (see Figure 9).

A total of seven participants took part in the user tests, each providing varying results. All participants were 20 years old or older and have experienced similar interactions before through other established websites. The following sections will give a summary of all the prototypes results, going over how the participants navigated the specific interactions and what their thoughts were. In addition, since the project took research through design approach it will also elaborate how each prototype evolved into one another.

4.4 First Prototype: Time-bounded Reading

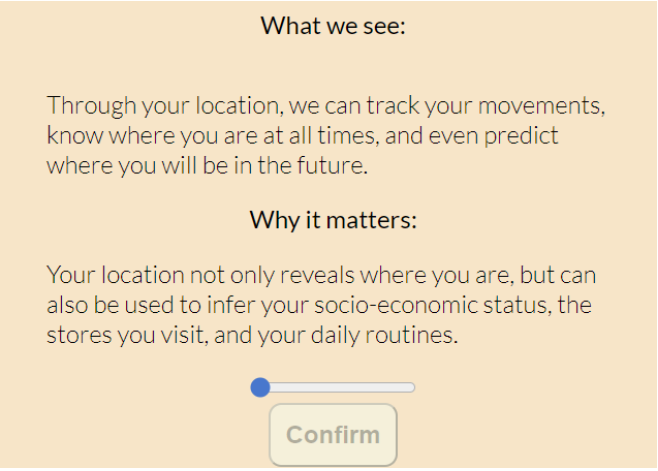


Figure 10. Brief description about the potential concerns that might occur when accepting.

The approach for the first prototype was inspired by Joshi's (2023) work done on time-bounded reading. As mentioned previously, their project revolved around keeping the user engaged through forced reading where the user cannot scroll through. The main concern for the prototype was the inclusion of a description. The previously mentioned examples concerning conveying information was that dark patterns include an abundance of it, so this prototypes purpose was to condense that. As shown on figure 10, the text is discussing about a fake scenario about the organisation being able to see where their location is, and why it matters. Both sections try to include as much blatant information that does not leave any deceptive meaning to it. The main reason for this approach is to keep everything as brief as possible to counteract the information overload that these interactions involve. This



Figure 11. First and second step are where the user puts their cursor to have the paragraphs typed out. Once finished, the user may confirm their choice.

prototype follows a stepwise approach where each highlighted part by the cursor will slowly type out the letters within the paragraph (see Figure 11). As previously mentioned, this is meant to reinforce the idea of forced reading within information-bearing forms. Additionally, whereas the previously

mentioned project was about scrolling this prototype focuses on keeping the user engaged on each paragraph as it types out slowly. As well as the paragraph does not type itself out once the user has moved their cursor over the location to initiate it.

This prototype is more concerned about visual clarity for which it does not contain too much interaction. The purpose is to convey the information through simple steps that force the user to see the sentences typed out in real-time. Once the user has finished each paragraph, a confirm button with a slider input on top will appear. The user can not simply press confirm but rather needs to fully drag the slider in order to make the confirm button active. In hindsight, this did add a necessary step within the interaction that was not important, simply because the focus was on reading. That said, the button appears a little before the last paragraph is finished so it was added as a counter measure to not let the user press confirm as soon as it appears.

4.4.1 1st Prototype: Result

This prototype received favourable result as participants expressed the slower interaction produced better reflection (see Figure 12). That said, some of the participants were confused as to how to start engaging with the prototype as it was not clear enough where that you were supposed to hover over the icons before it typed out. One participant expressed whilst the information being received was clear, the interaction between the paragraphs were confusing at most. The forced reading aspect was a strong addition to the friction that helped make the users be more conscious about what it is trying to explain. All the participants stated clearly afterwards what the descriptions tried to explain which was about tracing location data. Likewise,

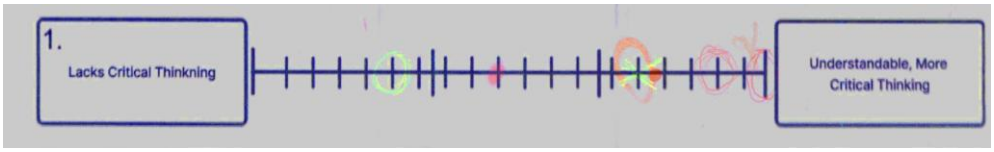


Figure 12. Summarized scale where all participants mark on the line has been layered on top of each other to get an estimate of the scale.

the brief explanation for it helped to elaborate it more since it does not involve any information overload. As previously stated, the interaction was not as feasible as expected. The forced engagement towards each paragraph was unclear as it was not properly explained to the user. In the prototype a cursor icon will slowly blink as to indicate where the user must place their mouse. Initially, a lot of the participants were not doing anything and expected the icon was about to perform something. It was not until explicitly stated what it was about that it clicked to the users. One of the participants even expressed how it should have been a hold on the input rather than a simple hover as to make it feel more determined by the user. Another point of contention was the inclusion of a slider before the user could press confirm. In hindsight, it felt quite like an unnecessary addition since at that point the user would have

finished each paragraph so adding more hinderance from the confirm is more of a nuisance. Some of the participants did not finish the slider thinking it was already over as it was never stated how it was connected to the confirm button. Although the interaction was not received favourably, the forced reading aspect did still produce good result concerning user safety and forming critical thinking towards the prototype. An aspect that was never realised during the development of the project was the notion of long-term fatigue concerning these interactions. One participant noted that while the interaction was good in forming critical thinking, long-term this would cause fatigue and become annoying as you are waiting for a countdown to the confirm button.

4.5 Second Prototype: Account Creation

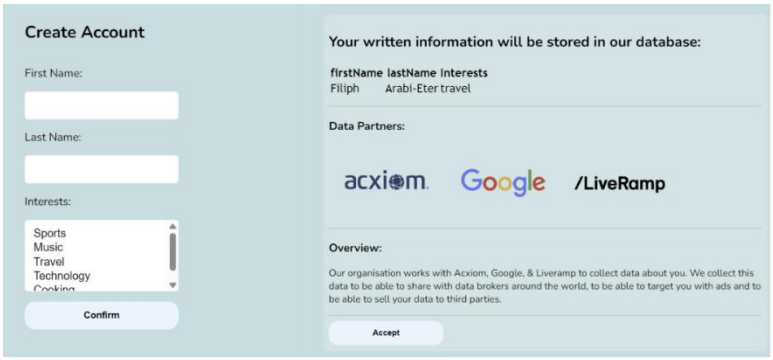


Figure 13. Image of the second prototype, left side contains the account creation, whereas the right side appears when the user has confirmed their account.

Once the first prototype was finished, ideas began form on how to iterate on less resistances and more on visual clarity. This prototype adds additional reading points to gauge the measurement of when something information is too much (see Figure 13). The point of the interaction is to mimic the typical account creation process many websites have. The user is only required to input their first name, last name, and select one interest from a predefined list. This design features fewer interactions that are inherently simple, consisting only of basic buttons and two text fields. Whereas the last the prototype was text-based, this one includes logos and outputs the users' written texts in a separate text field. Once the user has confirmed their account that data will be shown as feedback to the user within the form itself, highlighting which variable their data will be stored in. This design was inspired by established database structures that utilize structured query languages like SQL (Structured Query Language). Query languages utilize this structure to store data within a table, like a spreadsheet, where information is organized for easy retrieval. For instance, if a user provides

three data points such as a first name, last name, and an interest, each variable occupies its own column, while each individual entry constitutes a distinct row. This systematic organization allows for efficient filtering, where their information remains structured and accessible. In hindsight, it was difficult to convey that through this design since the data within the prototypes table will clear itself each time it is closed. This meant that the table only contained one user each time when it was created, making the

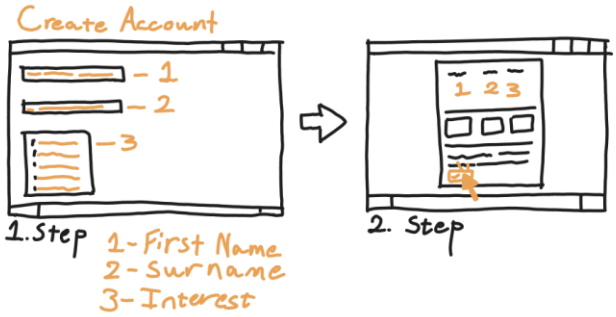


Figure 14. First step has the user typing out their information whilst choosing one interest. Second step is where a form appears with their written credentials and showcases a bunch of logos of which the website works with.

visuals of the table less predominant. The point of the project was not to store anyone's information when they created an account to it was difficult to assess how to share that data with users (see Figure 14).

Another prominent visual that appears right under the written credentials is the inclusion of showing fake data partners. As mentioned previously, many websites dictate their data collection through the policies of other third-parties. That said, in most cases they are never shown or disclosed. The logos included are Acxiom, Google, and LiveRamp, three of the largest data collectors (Pinchot et al. 2018). The reason to include this was due to adding more visual context rather than text-based. As a result, it leads to a more imposing design in which includes the actual organisations that oversee someone's data rather than keeping it obfuscated behind text. While cookie banners typically obscure the specific third parties involved, this prototype explicitly includes them to emphasize the vast, often unknown scale of data sharing operations. Afterwards there is additional text included to provide a more detailed description in how the prototypes fake organisations work accordingly with these third-parties. The prototype is meant to form a fictious visual that these are the potential third-party organisation you could share to, however being more direct and sharing that openly with the user. It is meant to highlight as stated previously by Bajwa & Meem (2024) regarding the necessity of showcasing a website's underlying mechanisms. By explicitly highlighting exactly where and how user information is stored, the prototype works to reduce the lack of transparency inherent in many modern web structures. That said, conveying this information presents a challenge, the

naming conventions of traditional programming are unfamiliar to most users. Consequently, the prototype's message might become obscured, as users may perceive the repeated display of their own information as merely repetitive rather than a transparent look at data storage.

4.5.1 2nd Prototype: Result

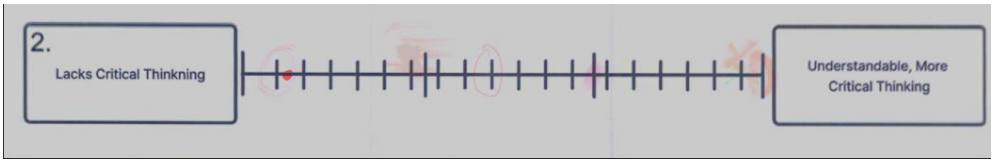


Figure 15. Summarized scale where all participants mark on the line has been layered on top of each other to get an estimate of the scale.

Building on the visual clarity of the first prototype, this second iteration shifts its focus and adds more visuals as the primary way to convey information, instead of primarily text. With the inclusion of logos, it was meant to inform the user where their information is being store. Within the scales it was a mix, either it received good critical thinking or the opposite with only one participant being in between (see Figure 15). The indecisive opinions mainly stem from it not having any resistances within the interaction. This prototype lacked any friction as the participants could easily pass this one without any attention to the messaging.

According to one participant, the inclusion of logos made the process feel more significant; they explained that seeing their personal information placed directly alongside these recognizable organizations felt more imposing. In addition, seeing what you typed being received back as a signalling towards what you are sending your information to do also receive mixed reception. Some of the participants did not really notice it, however, those who did explained how they it felt more negative. One participant noted that the logos felt significantly more imposing, causing them to question the potential outcomes. While there is a brief description underneath everything, it was not properly conveyed enough with the inclusion of the logos which made participant feel clueless about what it entails. The lack of information overloading was received as positive and no one expressed how everything felt bloated or too much. Likewise, both the first and second prototypes contained good visual clarity, however, this prototype did not have any thoughtful friction within it which ultimately led to the mixed reception. The main takeaway from the participants opinions is that there should have existed additional information to accompany the logos, in addition to adding more friction to make users less hastily skimming through the interaction.

4.6 Third Prototype: Complete the Sentence

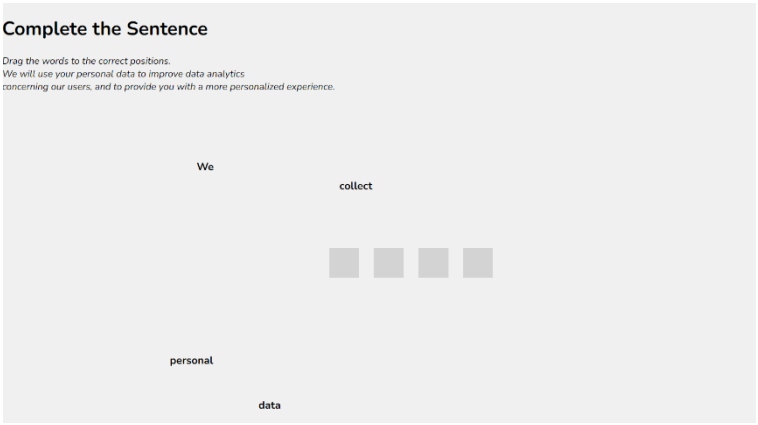


Figure 16. Third prototype where the user must drag the words scattered across the screen to complete the sentence.

Since the second prototype was easily skimmed through by most participants, attention shifted more towards additional resistances in the interaction to be made. Along each prototype the explanation became less cohesive and instead focused more on the interaction side of things. In this scenario it is about formulating the sentence yourself instead of reading it. It was meant to make the user conscious of the words that are being displayed. The reason for this choice was to go back to the first prototype on how to convey information in an effective manner. The third prototype follows a selection of four words that form the sentence “We collect personal data.” As seen on Figure 16, the words are initially scattered across the screen with no indication as to what its sentence is. There is a brief description within the top left corner explaining the task and its concept. The purpose of this prototype was to

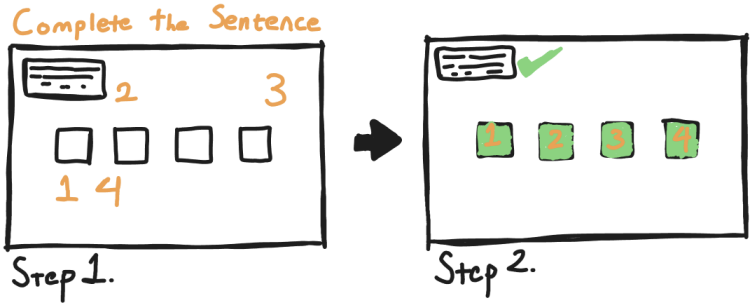


Figure 17. Step 1 shows how the words will randomly scatter themselves, and step 2 is where the words are correctly placed forming the sentence.

exaggerate the interaction compared to the last two where instead of merely presenting text to be read, it requires users to physically construct the sentences themselves by actively engaging with individual words (see Figure 17). By completing the sentence, the user is automatically shown a green background signify it is done with an accompanying text stating it. Whilst

brief, the description was not a focus of this prototype and instead the interaction. It follows similar principles with the first prototype, wherein it is exploring different ways to convey information through different means of engagement. Whereas the previous prototype explored reading through time-bounded text, this one pertains itself to having the user actively type it out themselves. Initially, the prototype was meant to be a mix between the first and second, where the typed credentials of the user would be put within the sentence; however, this was removed due to technical issues pertaining to the code. In essence, it caters itself to the interaction as it does not have that much information about data collecting nor data brokerage.

The thought process behind this interaction was inspired by CAPTCHA's, where users are evaluated with various puzzle-like interactions to be able to highlight that they are human and not a spambot. As previously mentioned by Cox (2016), they compared design friction to video games and how they usually involve multiple steps. By default, CAPTCHA's inherently have design friction in mind, however, in their case it is meant as a method for preventing spambots. That said, this prototype's interaction revolves again about conveying information through brief descriptions and engagement that requires users' attention. The words chosen for the sentence were purposefully blunt to be as apparent as possible. The word's 'data' and 'collect' were chosen specifically as to make the sentence as obvious as possible. Once the user selects the right word within one of the boxes, they will light green signifying it is within the right location. After all the words are within the right position the whole website turns green with text stating they have completed the task (see Figure 18). Though it may appear quite simple,



Figure 18. How the website looks once the sentence has been worded correctly.

the point is to get the feedback across and then brief the participant since the main question once it is finished is to gather thoughts and feedback around the interaction.

Regardless, due to technical constraints, the main design objective incorporating the user's own written input proved difficult to implement because of how the program was structured. Because of this, the sentence was hardcoded.

4.6.1 3rd Prototype: Result

This prototype when user tested received mixed results (see Figure 19). It is important to note that everyone easily completed the sentence and there was no one struggling to complete the task. Everything clicked quite fast for every participant with even the sentence being seen as obvious which no one had trouble figuring out. Although this one received mixed results after being

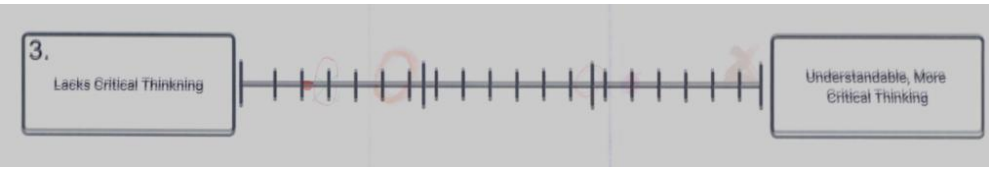


Figure 19. Summarized scale where all participants mark on the line has been layered on top of each other to get an estimate of the scale.

finished, participants stated that it mainly pertained to the fact that the sentence is coming of as obvious that the interaction is more of a nuisance. One participant noted that whilst the interaction was interesting as to letting the user figure out the statement, it should have contained more information concerning how it works. Whilst the interaction seemed somewhat feasible when observing the participants, it fell short in describing properly for the user how the practice ultimately works. That said, this shift to more gimmicky interaction could potentially prove be beneficial if implemented more properly with the right information provided.

4.7 Fourth Prototype: Input Fleeing

While the third prototype introduced more gimmicky elements, the fourth prototype pushes this concept even further. This prototype leans more heavily towards gimmickier interaction as to exaggerate the design friction across the various prototypes. The variety of prototypes was designed to identify the specific point where an interaction becomes excessive or overly gimmicky, and this was made aware of when creating this one particularly. It was meant to be the most exaggerated one out of all of them as opposed to being simple just as a means of conveying information. As the nature of

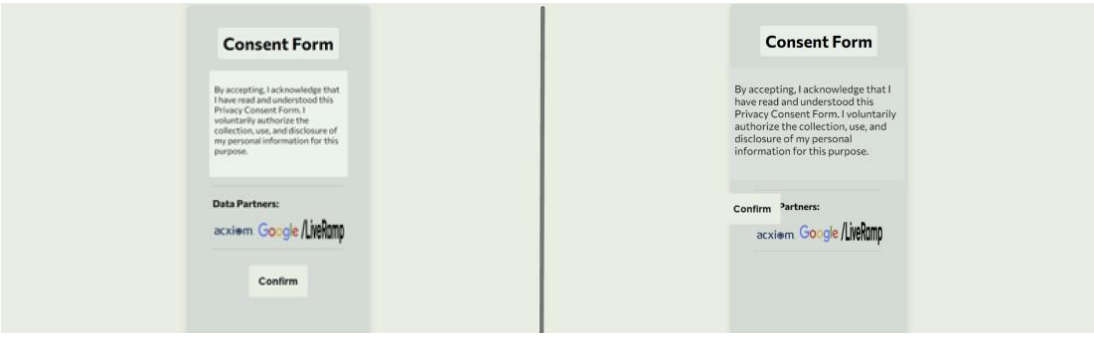


Figure 20. Left side is how it appears on initial setup, the right side showcases once the user hovers over the button, and it has moved away from the cursor.

normal cookie banners goes, they already have the preconceived notion which is nuisances. With that in mind, the line of which where this too much is an interesting exploration to have within these prototype as it can serve as a good inquiry from the future participants.

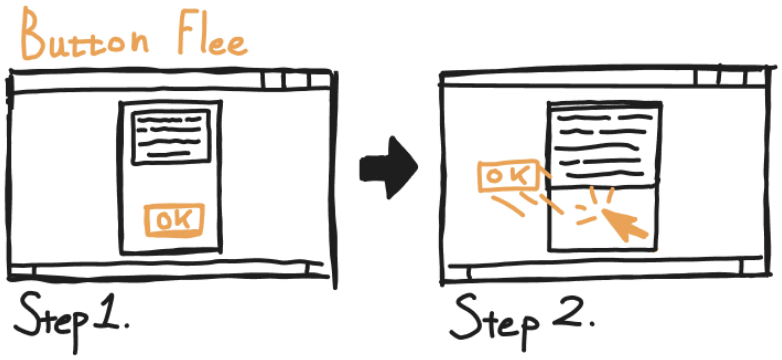


Figure 21. Demonstration on how the button reacts once the users' cursor hovers over the input, as well as the other elements becoming larger.

The process behind this prototype was to create something that came of as annoying to amplify the design friction. Hence, the idea behind having the confirm button do the opposite of what the user might expect became a prominent idea. Moreover, the purpose was to create something that subverts the users' expectations since the layout is quite typical and does not seem out of the ordinary. The layout of the website is meant to come off as ordinary and not form any initial expectations. However, as soon as the user interacts with the 'confirm' button it quickly moves away from the cursor position (see Figure 21. Consequently, the subversion is quickly apparent, and the user must chase the confirm button with the other elements on the website reacting every time as a response. Eventually, once the button has been chased around enough it will settle back to its initial position on the screen making it possible to press confirm.

Out of every prototype thus far, this one has the most exaggerated features and is the one made with being a gimmick in mind. Initially, the idea was to create something that just merely subverted the users' expectation briefly and quickly diminished itself. However, since a lot of the previous prototypes are already quite clear in their visuals and rigid, this prototype served as the one where it was merely done solely for experimental value. As the work progressed each became instinctively more focused on creating new interaction to explore, so it felt most suitable to continue that narrative and create an additional prototype meant to be hyperbolic as possible. Visually it shares similarities between the second prototype, with its inclusion of the same logos. That said, as to exaggerate the hyperbolic message of the prototype every element starts to violently shake in accordance with the user hovering over the input.

4.7.1 4th Prototype: Result

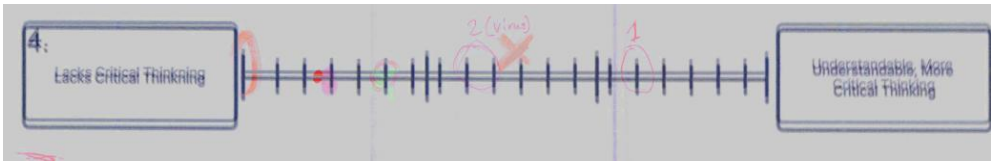


Figure 22. Summarized scale where all participants mark on the line has been layered on top of each other to get an estimate of the scale.

Continuing the gimmicky side of interactions, this prototype is by far the most and this was reflected a lot by the participants. Out of every prototype this one received the worst results within the scale and was expressed as the most annoying so far (see Figure 22). When observing the participants most of them did not finish the task at hand, which led to them feeling confused as to when they would be able to press confirm. Moreover, the constant chasing around the was perceived as frustrating by the participant with one making the comparison of a cat and mouse chase. Another participant made the comparison of an old computer virus from the early 2000's with its constant shaking and the input being misplaced. That said, because the confirm button was constantly pacing around the screen, the information on the screen lacked any presence which led to the participants forgetting what was stated. Additionally, the interaction lacked any critical thinking as the only focus on the input which made the prototype fall short in that department. Although it contained the most friction out of any prototype, it was perceived as too much and more of a frustration. One participant expressed that whilst the prototype was not feasible at all in terms of forming any reflection, if it was instead more similar to the second prototype it perhaps could have produced better result, but as it stands it did not generate any critical thinking with its friction.

4.8 Fifth Prototype: Smartphone Consent Form

As shown on Figure 23, the last prototype is the only one made explicitly for the smartphone. This was done as to differentiate the interactions and

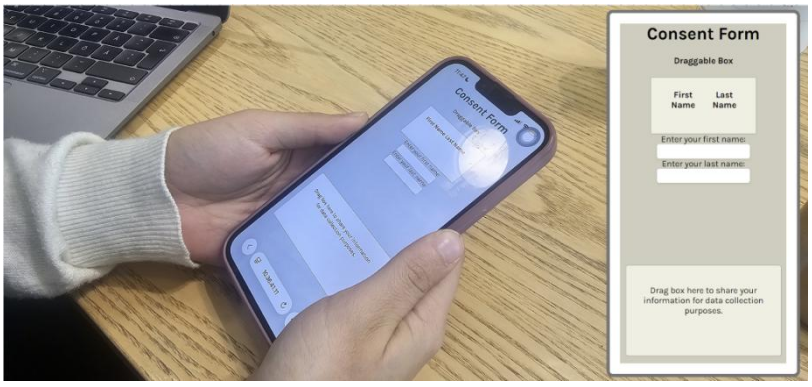


Figure 23. Picture of the prototype being used on the phone, right side shows a clearer visual on how it looks on the screen.

analyse any differences from the two. While the previous prototypes were designed with a desktop computer in mind to highlight specific mechanics, this mobile iteration was developed to explore whether the platform itself changes the users' perception. By shifting to a mobile context, the goal was to determine if the device's inherent touch-based nature could potentially produce another unique response within the context of consent forms.

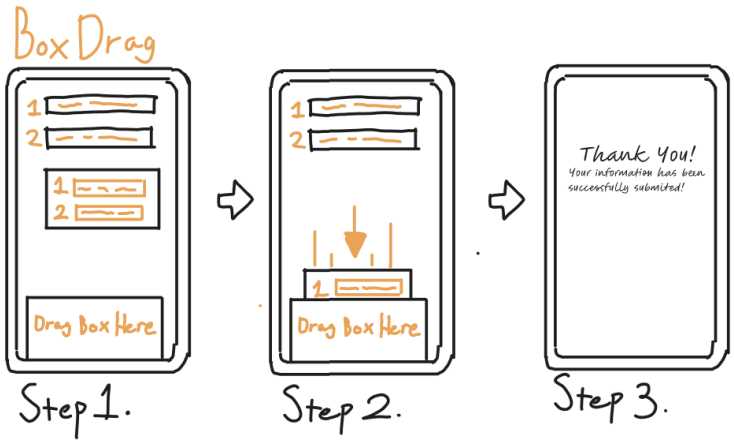


Figure 24. Step 1: The user must type in their information within the text fields, which later gets written as well within the box underneath. Step 2: User must drag the box with information below. Step 3: The user has successfully transferred their information.

The idea for the smartphone interaction was to create something where the user must physically send their information to the organisation, similar to how a consent form in real life might transpire. There was supposed to be a sense of willingness of sending the information, as the previous interviews indicated, users sometimes perform certain digital tasks lacking full awareness of the underlying data exchange. Thus, the smartphone serves as an interesting research tool for its possibility of making those specific interactions possible. Furthermore, given the heavy emphasis on desktop-based prototypes in the previous sections, it felt suitable to also explore smartphones since these types of interactions occur there as well.

As seen on Figure 24, the first step of the process is for the user to type out their first name and last name. As the user is typing out their information, another box on the screen is typing those same letters within its field, like the second prototype. In the end, it will end up showing their first and last name within that box and will function as the sender for the user to drag. Once everything has been written, the user must drag the box within the bottom of the screen to confirm their consent. If everything is done accordingly the screen will display a message for the user stating that they have finished the task. The user must have the fields within the box filled otherwise it cannot be submitted.

The idea for this prototype did not have any significant changes over the course of development, it remained quite steadfast. As mentioned previously concerning showing the mechanisms, this prototype was also meant to

inherit those choices. That said, rather than the previous approach this was to be where the user actively sees the choices being made secondary to what they are doing, as an active response. To ensure the interaction remained visible, the input box shown in Figure 17 was positioned above the text fields, this helped to prevent the smartphone keyboard obstructing its view. Ultimately, by comparison this prototype inherits a lot of the established tropes that have been thus far. That being, making the user perform tasks that cater towards a willing and making them aware of its actions. However, while none of the prototypes involves any choices to be made, this was an intentional decision. The primary objective was to explore how front-end design can be restructured to create an environment that is more transparent and user centric.

4.8.1 5th Prototype: Result

Because this prototype was the only one to include smartphone interaction, it helped to generate more unique results. In terms of an overview of every result, it was as well quite mixed within the scale (see Figure 25). Many participants mentioned that the interface felt most at home on a smartphone

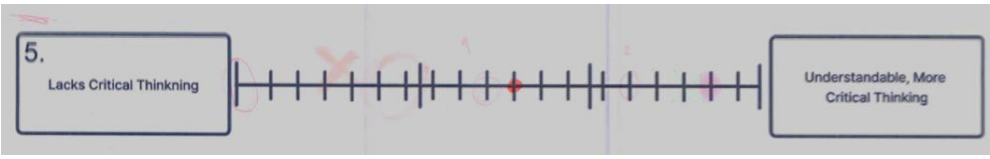


Figure 25. Summarized scale where all participants mark on the line has been layered on top of each other to get an estimate of the scale.

and would not have worked as well on a computer. This ease of use was evident in practice as everyone finished the task perfectly without a single hiccup. However, the mixed reception is difficult to assess as the participant themselves had trouble formulating their reasoning.

In general, while the interaction was perceived favourably, the reason it did not seem as reflective could be because the interaction bears too much focus. Since the interaction involved the user themselves purposefully sending their information, means the other things might have gone unnoticed. One participant expressed how the interaction felt like they were signing an envelope and sending it to the organization, which made it feel more physical. This was a common consensus throughout all participant, which was that there was more of a physical aspect pertaining to it. This stems from the fact that it was developed for the smartphone, had it been on a flatscreen it might have not produced the same result. Another participant mentioned that regardless of having to physically send the information that it should have instead been simpler, proposing the idea that holding would make them feel more conscious of their decision. Similarly, the previous prototype was too interaction focused, which this prototype seems to have an issue with as well. As evident by the participants, whilst the aspect of having your information being physically sent was received favourably, there still lacked critical

thinking. Additionally, there were issues regarding user friction, specifically, the interaction failed to ensure that participants read the alert message after the information was sent. Ultimately, the main takeaway from the results was that there is a clear unbalance between the friction and the visuals. When observing the participants with how fast they were proceeding through, the visuals when unnoticed which could be the reason as to it receiving mixed results.

4.9 Analysis & Identifying Themes

User testing revealed a significant mix in participant reception across the prototypes. The lack of consensus makes it difficult to assess the actual core problem. This could be because of the scale used being too vague and not really feasible as a result. In retrospect, it should have pertained to a better structure of question or another method for writing down thoughts. That said, the prototypes largely succeeded in their primary goal by adding friction to heighten user awareness. Most notably, while the friction achieved its goal of increasing mindfulness, it simultaneously had a lot of preconceived notions following them since they were seen as already of a nuisance. The user tests highlight a clear divergence in how users process friction. What one participant dismissed as a lack of depth, another embraced as a meaningful

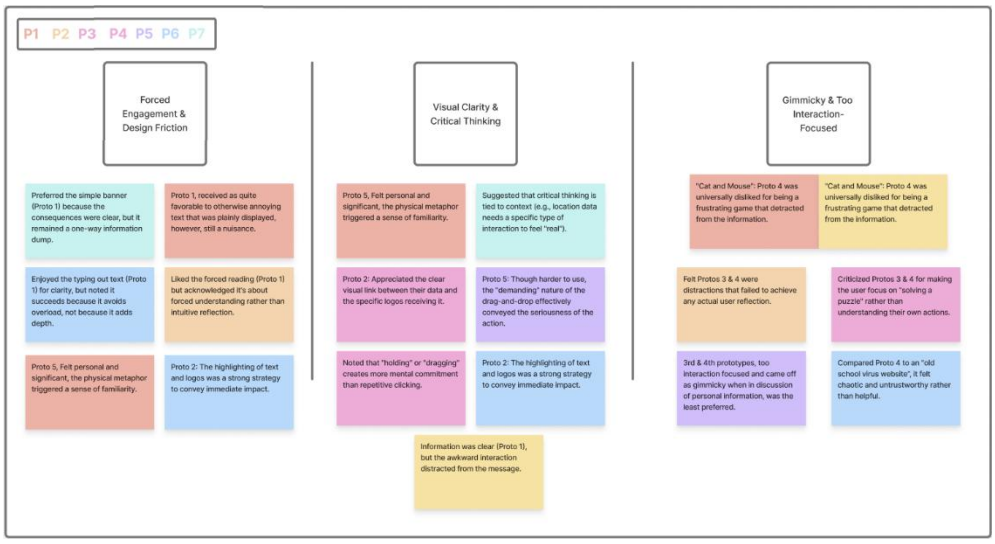


Figure 26. Notes taken during user-tests being condensed and summarized within an affinity diagram.

moment of pause. This indicates that it is difficult to create a single solution for counteracting dark patterns as it remains in a sensitive setting which in this case is personal data. As seen on figure 26, due to the amount of information from the participants, an affinity diagram was created as to easily connect the feedback to the themes. The three themes are: design friction & forced engagement, visual clarity & critical thinking, and gimmicky & too interaction focused. Suffice to say, these themes represent a common trope found throughout all the prototypes and highlight recurring feedback that all

of them pertain to. It is important to note the reason for choosing these themes stem from the information gathered from each user test. They represent an overall summary of each prototype being categorized into these themes. Despite some participants stating an interaction may be seen as a nuisance, in the context of design friction it should not be seen as a shortcoming but rather an opportunity where the information can be received through. Banners are already considered an annoyance, which in the case of friction is a great opportunity for exploring how it can rather be transformed into something more thoughtful. As mentioned previously by Mejtoft et al. (2019), they argued that design friction is meant as a way of hinderance, which will inevitably be met with some criticisms from users. In essence, there should still be a balance between thoughtful design frictions and clear visuals, hence, the inclusion of the theme 'gimmicky.

These prototypes were meant to question dark patterns on how friction can promote critical thinking, proving that front-end design can be used to potentially spark reflection instead of suppressing it. This is a direct response to the predatory design choices favoured for data brokers, who treat the front-end as a tool for data collecting.

5 Discussion

5.1 Thematic Analysis

5.1.1 Design Friction & Forced Engagement

This theme is primarily focused on summarizing to which prototype received favourable result of design friction. The first prototype received the most thought concerning it since it is restricted to time in order to complete the task. Despite previously mentioning the interaction lacking in intuitiveness it still holds some regards for its mandatory reflection. The second prototype lacks in this theme since participants skimmed through it quite quickly. In retrospect, the second prototype lacked in resistances and should have elaborated on the already established workings of the previous one. That said, it highlights how the design friction can occur through simple visuals that occur overtime. One aspect that was brought up by one participant is how future interactions could cause fatigue. They elaborate that the prototype now felt thoughtful but overtime it will be seen as more annoying since it is tied to a timer which cannot be completed until a certain limit is reached. This was initially never considered when creating the prototype, however, it may be an inevitability that occurs regardless of which design prevails the most. Another prototype that received unique feedback was the fifth smartphone prototype. Although its interaction may be trivial, the act of creating something that was familiar to a real act such as sending an envelope helped to establish

reflection as well. This is the only prototype in which was able to produce those thoughts as the other ones was never compared to anything else that was related to a physical action. Additionally, because this prototype was designed for smartphones, its success could be attributed to the physical quality of the interaction.

In the context of personal information, it is important for the front-end to ensure that despite an interaction being forced upon users; there is still potential for the design to formulate thoughtful reflection. Given the sensitive nature of personal data, the interface must aesthetically and functionally mirror the gravity of the interaction. Even when perceived as a nuisance, intentional friction serves as a necessary cognitive resistance, shifting users from passive clicking to more conscious decisions.

5.1.2 Visual Clarity & Critical Thinking

One of the most notable inputs received out of all five was the first and second, which relied on visual clarity to promote critical thinking. The first and second prototype was made with visual clarity in mind following Otton (2020) approach to the subject. As previously mentioned, they argued that aesthetics and usability can largely be determined by the perception of the user interface, which these prototypes follow accordingly by not overwhelming the user with information. Most notably the first prototype was expressed the most indicating that regardless of having to read, information bears a significant amount of purpose when conveying what their personal information could entail to. Design opportunities regarding a forced aspect of reading or visuals can prove potentially feasible in design frictions as they offer a clear sentiment towards the user whilst not being overbearing. Furthermore, utilizing friction to convey information must be balanced with visual clarity. As noted by participants, while certain prototypes were still perceived as a nuisance, their lack of information overload allowed for a higher degree of critical thinking. One participant feedback reveals that even when an interaction feels awkward, it can remain reflective if the underlying information is clear. This proves that good usability does not always have to be the goal when the objective is to force a conscious pause in the user's decision-making process.

Whilst the saying that the interaction was awkward or a nuisance the fact remains that the friction in this case worked, as they are not meant to be pleasant or easy to skim through. In addition, because the interaction was meant to be created simple does not persuade the fact that it was to be easy to get through. Which in the theme of critical thinking means that it helps to promote it. This indicates that regardless of user perception on the interaction if the participant can recall the information they received from the visuals, the friction in this case is only meant to enforce it with the interaction.

5.1.3 Gimmicky & Too interaction-focused

While including interaction-heavy prototypes may appear counterintuitive, they served as a critical baseline to identify where the engagement was too much of the focus leaving critical thinking out of the question. These themes were brought up the most during the third and fourth prototype, which were purposefully made that in mind. The fourth prototype especially was received the most as being too gimmicky and the participants expressed that quite a lot since most of them forget the information being shown. It highlights how even if design friction is meant to be a hinderance, there exists a gauge where it will be too much. Two participants mentioned that it was similar to a cat and mouse game, which in the context of critical thinking is far to derived from its point. The main takeaway from each interaction should be that the user is understanding of what their personal information may entail not to persuade them from the primary subject which is what dark patterns are trying to do. The balance in which interaction and critical thinking occur does not happen when it overbears the other, as evident by the third and fourth. Although the third prototype required users to manually construct sentences detailing data consequences, it failed to elicit meaningful reflection. Instead of engaging with the content, participants approached the task as a puzzle solver which gamifies the process. This shift suggests that when interaction is too high, the task of completing it overrides the user's ability to process the actual information being conveyed towards them. Ultimately, this particular theme should not be as desired from designers and requires significant thoughts towards.

5.2 Result

The research questions regarding how front-end interface friction creates user awareness, and if slower interactions can promote critical thinking proves to be highly complex. This is in part due to the subjective nature of the user experiences. However, regardless of if the friction is received as poorly does not mean it should not be dismissed as a flaw necessarily. This is mainly because of the context in which it serves, which in this projects case is about personal information. If the participant can recall what was being conveyed in the information from the prototype, does that mean the friction has failed? Friction is not meant to form a sense of enjoyment but instead help acknowledge the users' action and help create better awareness amongst the interaction. Despite some of the prototypes being perceived as a nuisance, they still were able in some aspect to form a reflective experience. By taking the pre-existing interactions of dark patterns and rethinking them, it can transform it into a more structured obstacle that forces users to acknowledge what actions entail. Rather than merely replicating existing mechanics for the sake of repetition, this approach requires reimagining interactions to be more intentional, shifting users away from hasty decisions.

Findings from the design process reveal that slower interaction has potential within front-end interfaces but requires a significant balance on how to approach it. Its effectiveness depends on the balance between visual clarity and friction. For example, the previously discussed time-bounded prototype demonstrates how an alignment of visuals and friction can coexist effectively. This synergy shows clear potential for slowing down user pacing and encouraging more deliberate decision-making.

5.3 Critical Reflection

Arguably the biggest challenge for the project was assessing the scale, as it does not contain any concrete ratings. The wording 'Lack of critical thinking' to 'Understandable, More critical thinking' does not contain any thoughtful consideration to the prototypes. It comes off as arbitrary choices with no sense of accurately measuring each prototype. In retrospect, quantitative methods should probably not have been included as it did not form any feasible results, instead the qualitative methods from the field notes produced more interesting findings. Moreover, due to the fact that a scale was introduced, it would affect the thematic analysis. While this study intentionally adopted a deductive approach, employing a different qualitative method for each prototype could have facilitated an inductive analysis. Transitioning to an inductive framework would have allowed for unique emergent themes to surface directly from the participants rather than fitting field notes into predefined categories.

Although the design process for the most part was quite smooth, the research through design approach could perhaps been rethought of. That is to say, the prototypes became more of a form of inquiry rather than iterating on them. As each prototype was distinctly made from each other, it could have possibly produced more of a well-defined version of what design friction can be. While this project effectively maps the exploratory landscape of design friction, it lacks a precise definition of its potential, leaving room for future work to establish concrete implementation. With this in mind, the inductive approach could have then perhaps generated a unique response to contribute for design friction.

5.4 Future Work & Suggestions

Given the variety of opinions regarding this topic from participants, it still answered some parts of the research question, which in this case is that it can promote reflection. However, there is still more potential for this topic to investigate further, which could be to explore other forms of friction. As was previously discussed, friction is not only upheld to solely be confined to very simple stops. Friction can occur under many different circumstances such as off-putting colouring or unorthodox placements within the user interface. There are no set boundaries for how to approach design friction, Benedetti & Mauri (2023) mentioned how friction is meant to recontextualize existing

data and challenge its initial design. These recontextualizations are to make the goal of the user interface be more challenging in ways that are not typical for traditional user interfaces. The approaches that were taken so far have proved to be somewhat effective in those regards, however, there is a line in which is important for designers to understand. Whilst friction has the potential to be made into something reflective, it can still take over the interaction and become the focus depending on how much resistances is being applied. As evident by the prototypes, friction should not bear too much focus on banners. That said, regardless of what users may perceive the nuisances within the interaction, friction should still be seen as a positive in those regards.

6 Conclusions

This thesis investigates the potential for design friction within front-end interfaces to promote critical thinking. Despite there existing a lot of back-end concerns with collecting of personal information, front-end should still be held responsible due to the many malicious practices that occur on various website. Terms such as ‘dark patterns’ have been used to describe such cases where the design is deceptively done. It is meant to describe the ongoing design practices within web interfaces that favour the organisations rather than the users. Moreover, dark patterns may overwhelm users with information or persuade them into doing something of which they are not fully aware. Adopting a Research through Design (RtD) methodology, the project utilizes the development and testing of prototypes as the primary method for gathering information and doing research. A total of five prototype was made, and each were meant to gauge where the limit is concerning user engagement and critical thinking. In essence, user tests reveals that regardless of reception concerning each prototype, friction has potential to make users aware of what their actions could entail. Despite participants viewing some interaction models as an annoyance, the intentional friction succeeded in heightening awareness.

A review of existing literature suggests that design friction is traditionally defined as a hindrance that contains resistances within the interactions, which inevitably will lead to some user frustration. Participant feedback indicates that friction is a double-edged sword. While it can successfully disrupt mindless interactions, there exists a fine line where the interaction bears too much and makes the user forget what their action can cause. For information to be received towards the user, the design must have some active engagement with the information to ensure that the user can reflect upon their decisions and not skim through it easily. It is important for designers to understand the balance of when friction is too much. While

design friction offers an opportunity for users to reflect upon certain details, it can as well make the interaction bear too much focus regardless of the context in which it serves.

One notable aspect which was not explored properly was providing better context for back-end data collection into the front-end through visual means. The prototypes towards the end focused heavily on the interaction side of things without taking visuals that much into consideration. As friction has the potential to be implemented through circumstances, opportunities exist to integrate these safeguards directly with data structures. Designers can then surface these structural constraints through the front-end interface to increase user awareness.

7 Acknowledgement

I would like to extend my gratitude to my supervisor Per Linde for all the help that was given in this project, their guidance was of great assistance. Thanks to all the participants that took part in this research, as well those who took their time to user test the prototypes and provide valuable feedback.

Last but not least, thanks to my close friends Valdemar, Lina, and Roza who managed to put up with me during the most stressful times.

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9 Appendix

MALMÖ UNIVERSITY

Information letter Appendix 1

Project title: Designing for Awareness: Applying Friction in Dark Patterns	**Date:**
Study manager: Filiph Arabi-Eter **Your e-mail** filiph2002@gmail.com	**Studying at Malmö University,** **Faculty of Culture and Society,** **S-205 06 Malmö,** **Phone +46 40 665 70 00** **Education:** Interaction Design **Level:** Bachelor

The following information should be included:

– Presentation:
Filiph Arabi-Eter, studying Interaction Design at Malmö University, Bachelor level.

– Aim:
The aim behind this project is to research and evaluate user engagement regarding deceptive design tactics and how to counteract those practices.

– Implementation:
Interviews will follow a semi-structured discussion that will be around 10-15 minutes. As there will be several distinct prototypes made there can occur more than one user tests, with each being quite brief. Additionally, there will be a small think-aloud session afterwards where participants can express their opinions regarding the prototype.

– The voluntary:
Your participation is completely voluntary and can be stopped at any moment with no explanation required.

– The confidentiality:
We strive to guarantee confidentiality in the study in that no unauthorised person may have access to the material. The material is stored so that it is only accessible for the individual or individuals leading the study. In the reporting of results in the form of a degree project paper at Malmö University/or in another form of publication, the respondents will be unidentifiable, and it will not be possible to link the results to individuals.

GDPR
The thesis will be published through DiVA portal
https://diva-portal.org

You are hereby asked to take part in this study.

MALMÖ UNIVERSITY

Informed consent Appendix 2

Project title: Designing for Awareness: Applying Friction in Dark Patterns	**Date:**
Study manager: Filiph Arabi-Eter **Your E-mail:** filiph2002@gmail.com	**Studying at Malmö University,** **Faculty of Culture and Society,** **S-205 06 Malmö** **Phone +46 40 665 70 00** **Education:** Interaction Design **Level:** Bachelor

I have been verbally informed about the study and read the accompanying written information. I am aware that my participation is voluntary, and that I, at any time and without explanation, can withdraw my participation. The person/-s leading the study will strive to guarantee confidentiality in that no unauthorized person may have access to the material. The gathered material will be stored properly and used for research purposes only.

I hereby submit my consent to participate in the above survey:

Date: ……..

Participant's signature: ……